



Thin film LiNbO_3 surface preparation using SC-1

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Abstract: This study investigates the use of Standard Cleaning 1 (SC-1), a routine treatment in silicon micro-/nano-fabrication, as a step in the processing of photonic structures and waveguides on thin films of lithium niobate on insulator (LNOI). We investigate the use of SC-1 for surface preparation and as a cleaning method to remove residual byproducts after ICP etching, evaluating its impact on etching dynamics and the quality of the LiNbO_3 thin film. Our results demonstrate that SC-1 cleaning removes a surface layer of LiNbO_3 characterized by higher oxygen vacancies. These oxygen vacancies accelerate the etching process and lead to faster under-etching of Si_3N_4 capping layers on LiNbO_3 , causing pattern delamination, particularly at sub-micron scales where under-etching is more detrimental. Finally, we show that carrying out a SC-1 treatment preliminary to Si_3N_4 capping layer deposition enables the fabrication of Si_3N_4 gratings with sub-100 nm width on LiNbO_3 thin films.

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1. Introduction

Lithium Niobate (LiNbO_3) [1] is a material of exceptional importance in photonics and optoelectronics, thanks to its wide transparency window (0.4 – 5 μm) [2], and, more importantly, its strong nonlinear optical properties [3] including an attractive electro-optic response [4]. These characteristics have made it indispensable for a wide variety of photonic components, such as electro-optic modulators [5], frequency converters [6], optical parametric oscillators [7], and integrated waveguides [8].

Moreover, the emergence of LiNbO_3 on insulator (LNOI) [9], also known as thin film lithium niobate (TFLN) on insulator has further transformed the field of integrated photonics [10]. High-quality sub-micron LiNbO_3 films, typically fabricated on silicon, silica or other substrates, offer strong optical confinement and low propagation losses. As a result, LNOI has significantly broadened the scope of LiNbO_3 applications, particularly in areas such as integrated quantum photonics, high-speed signal modulation, frequency comb generation, and compact optical sensing [11].

A key challenge for the fabrication of these photonic structures is the inertness of lithium niobate, hindering direct etching of narrow ridge waveguides and grating couplers into lithium niobate. For direct etching of LiNbO_3 , many different processes have been used, the prevailing one being based on argon (Ar) plasma etching, and most if not all of them rely on the so-called "Standard Cleaning step 1" (SC-1) [12] to remove the etching products redeposited on sidewalls [13]. Another widely used approach is to cap lithium niobate with a silicon nitride layer into which the various lateral features are etched using plasma etching. In that case also, SC-1 is subsequently used to clean the residues coming from the overetching into the lithium niobate thin film.

As it will be shown in section 3, working on the latter approach and focusing on narrow ridge waveguides and sub-micron gratings, we observed that narrow ridge waveguides and sub-micron gratings tend to exfoliate during SC-1 cleaning. In this work, we delve into the details of this

essential cleaning step, rarely detailed in the literature. A comprehensive investigation into the chemical interactions between LiNbO_3 and the cleaning agents is presented. We propose and demonstrate an improved process-flow harnessing SC-1 for both initial surface restoration of LNOI and subsequent cleaning of dry etching residues, that, in turn, enables reliable and repeatable patterning of features down to sub 100 nm critical dimensions.

The paper is organized as follows:

Section 2 presents the different samples used during this study. Section 3 shows the exfoliation of sub-micron gratings after SC-1 cleaning. Section 4 is a study of the SC-1 cleaning dynamics while section 5 presents structural and topological impact of SC-1 cleaning. Section 7 presents our improved fabrication protocol including surface restoration and residue cleaning and the successful fabrication of grating structures with sub-100 nm features.

2. Samples description and initial fabrication protocol

To support our study, several samples were fabricated and analyzed. These $8 \times 12 \text{ mm}^2$ samples were diced from a 4-inch wafer purchased from NanoLN [14]. The wafer consisted of an X-cut LiNbO_3 thin film ($\sim 300 \text{ nm}$ thick), bonded to bulk LiNbO_3 with a $2\text{-}\mu\text{m}$ PECVD SiO_2 buffer oxide (BOX), onto an SiO_2 substrate. Such stacks are commonly used for strip-loaded waveguides on TFLN [15–17] or for sub-wavelength optical filters [18].

As shown in Fig. 1, the vertical stack of our structures contains a $500 \mu\text{m}$ thick SiO_2 substrate, a $2 \mu\text{m}$ thick SiO_2 spacer, a 300 nm thick LiNbO_3 thin film and a 70 nm thick Si_3N_4 top layer into which are etched various gratings of pitch p and pattern width w .

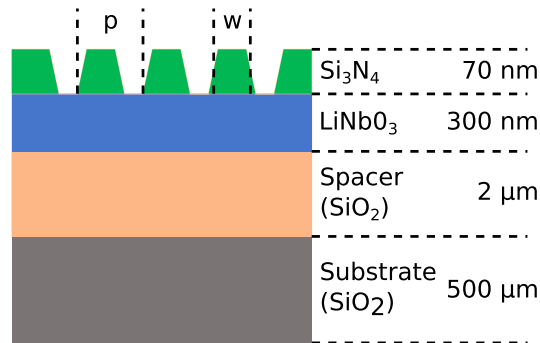


Fig. 1. Vertical stack of the structures under study incorporating Si_3N_4 gratings of pitch p and pattern width w on a LNOI platform.

Broadly speaking, the fabrication of waveguides and gratings on LiNbO_3 thin films can, in general, be considered as a relatively straightforward process, typically involving a sequence of dielectric deposition, lithographic patterning, and dry/wet etching [18].

An overview of the standard fabrication process-flow is presented below.

1. Cleaning:

Acetone/IPA : 5 minutes each

Plasma O_2 : 5 minutes, 1000 sccm O_2 , 1 mbar, 400 W RF power

2. Si_3N_4 deposition

Process gases: N_2 at 26 sccm and SiH_4 at 26.5 sccm

RF power: 500 W

Chamber pressure: 7.5×10^{-9} Torr

3. Patterning

4. Si₃N₄ Etching

Chiller temperature: 10 °C

Wafer electrode temperature: 20 °C

Chamber pressure: 35 mTorr (with 0.1 Pa offset correction)

Process gases: CHF₃ at 50 sccm, O₂ at 3 sccm,

RF power: 100 W source power, matching auto

Wafer backside cooling: 800 Pa

5. Resist stripping

Samples are first cleaned with acetone and isopropanol (IPA) for 5 minutes each, followed by an oxygen plasma treatment to remove residual organic contaminants. Si₃N₄ deposition is then performed using ICP-PECVD technique. Subsequently, the sample is patterned using electron beam lithography (EBL, Raith 150) with 130 nm-thick PMMA as the masking resist. To mitigate charging effects during exposure, PMMA is capped with Electra 92, a conductive resist. EBL is carried out with a 20 kV acceleration voltage, a nominal dose of 220 $\mu\text{C}/\text{cm}^2$. After exposure, Electra 92 is removed in deionized water bath (1 minute), and PMMA is developed using MIBK:IPA (1:3) solution for 45 seconds. The transfer of the resist pattern into the subjacent Si₃N₄ layer is performed using a ICP-RIE process on a SI 500 DRIE system. The resist excess and residues are then removed using an acetone/IPA bath followed by a 5-minute oxygen plasma ashing, a common process widely described in the literature [19].

Using this process-flow as a starting point, various samples were fabricated and are listed in Table 1 which provides the key specific treatments, and the purpose of each sample.

Table 1. The samples used in this work, along with their treatments and experimental purposes.

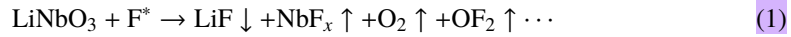
Sample	Treatment	Purpose
A	Acetone/IPA	Grating fabrication without surface preparation
B	SC-1/Plasma O ₂	SC-1 Etching dynamics & roughness measurement
C	SC-1/Plasma O ₂	
D	Acetone/IPA	Surface topology
E	Plasma O ₂	XPS study
F	SC-1	XPS study
G	SC-1 prep	XPS study
		Grating fabrication with surface preparation

Sample A was dedicated to the fabrication of silicon nitride gratings with sub-micron features, without initial surface preparation, demonstrating the delamination issue of small features that motivated this study. Sample B was used to study the etching dynamics of the SC-1 cleaning. With Sample C, we observed the role of SC-1 in roughness reduction and topology preservation. Samples D to F were used to track elemental composition evolution of the LNOI surface using X-ray photoelectron spectroscopy (XPS). Last, Sample G is a twin of Sample A but with the improved fabrication protocol resulting from this study, including in particular an initial surface preparation and restoration using an SC-1 treatment.

3. Initial observations

When using the process-flow presented above, to ensure controlled waveguide confinement or metasurface optical response, it is customary to remove in full the dielectric material. A slight

overetch is therefore performed [20], capitalizing on the lithium niobate chemical inertness to introduce a surface etch-stop and thereby limit aspect-ratio-dependent etching effects. In turn, this overetch induces some redeposition on the exposed top surfaces and sidewalls of etching by-products, like crystalline lithium fluoride (LiF) that is created following the reaction below [21–23]:



This phenomenon is illustrated in Fig. 2(Left), which presents a Scanning Electron Microscope (SEM) image of a sample (Sample A, see section 2) showing the Si_3N_4 grating patterns after etching and mask removal.

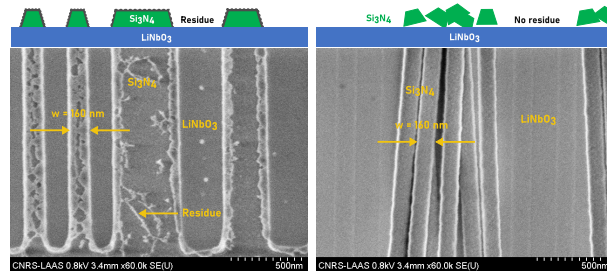


Fig. 2. (Left) SEM image of Si_3N_4 gratings etched using fluorine-based ICP etching, shown after mask removal. The minimum line width is $w = 160$ nm. Residual etch by-products are still visible on the surface. A schematic above the image illustrates the grating layout. (Right) SEM image of the same sample after cleaning to remove the by-products, revealing grating collapse and delamination. The coalescence of features and shadowed regions in the background indicate areas where the patterns have peeled off, suggesting poor adhesion between the Si_3N_4 and the underlying LiNbO_3 layer.

Residual etch by-products must be thoroughly removed, as their presence can result in a rough LiNbO_3 surface that is unsuitable for optical applications [24]. This can lead to increased optical losses and adversely affect subsequent film depositions, such as anti-reflection (AR) coatings [25].

In the literature, the SC-1 step (Standard Clean 1) of the RCA cleaning procedure is used to remove such residues [26–28]. When we introduced SC-1 cleaning as a final step in our processing, we observed that whilst effective in removing residues and other organic contaminants, the SC-1 treatment also led to the peeling of the grating structures, as shown in Fig. 2 (Right). The image clearly shows that the grating lines have lifted off and coalesced, the weak-contrast marks being indicative of the areas where from these features detached. At this stage, we would like to emphasize that, following the residual by-products cleaning step, we consistently observe exfoliation of the dielectric gratings for a wide range of sub-micron widths (and in particular, here, for $w = 160$ nm), and this across multiple similarly-processed samples. It is also worth highlighting that, as can be seen in Fig. 3, the larger features akin to ridge waveguide widths – with $w \geq 500$ nm – were unaffected, justifying, at least part, why this issue has been largely overlooked to-date in the literature.

Gaining a better understanding of the role played by the SC-1 treatment and its action on the LiNbO_3 surface was thus deemed essential to solve this delamination challenge and motivated the work reported hereafter.

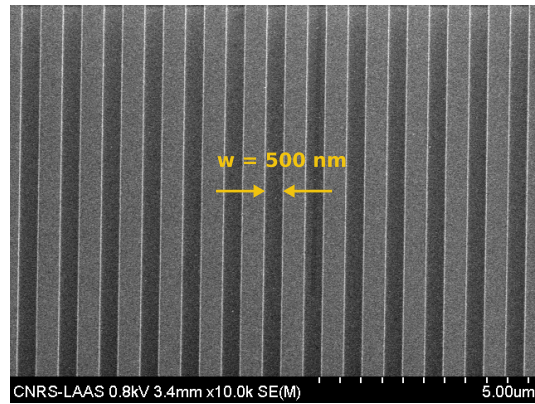


Fig. 3. Intact large-pitch gratings ($p = 1000$ nm, $w = 500$ nm) on Sample A – without SC-1 surface preparation.

4. Standard clean 1: etching dynamics

SC-1 solution, consisting of deionized water (DI), hydrogen peroxide (H_2O_2), and ammonium hydroxide (NH_4OH), is widely used in microfabrication, particularly for cleaning and wet etching processes. In silicon electronics and photonics, SC-1 is employed to clean surfaces before critical processing steps, such as oxidation, deposition, and lithography, to ensure that surfaces are free of organic contaminants, metal residues, and particles.

In this section, we study the dynamic of SC-1 interaction with the LiNbO_3 top surface using a dedicated sample – Sample B – where no dielectric layer is deposited on the LNOI.

The SC-1 solution was prepared by mixing NH_4OH (28-30 % w/w in H_2O), H_2O_2 (30 % w/w in H_2O), and DI in a 1:1:5 ratio. First, 400 mL of DI was heated to 65°C , followed by the addition of 80 mL of NH_4OH . Then, 80 mL of H_2O_2 was added. The chemicals were added sequentially to prevent a change in the concentration (during the time required to achieve the desired temperature), especially of NH_4OH , as it has a low boiling point of 38°C [29]. For short treatment times, using a covered container is sufficient to minimize the loss of volatile species; however, for longer processing durations, equipment such as sealed vessels or reflux condensers may be employed to further reduce volatility and maintain solution reproducibility.

The addition of the chemicals caused the temperature to drop by few degrees, so the solution was allowed to stabilize back to 65°C before introducing the Sample B. Stirring at 625 rpm ensured uniform mixing of the solution.

The thickness of the LiNbO_3 thin film was monitored using spatially resolved ellipsometry (Accurion EP4) at regular intervals. Spectra were recorded over the 360-900 nm wavelength range at 55 spectral points and subsequently fitted using tabulated optical constants for lithium niobate [30] and SiO_2 [31]. For Si_3N_4 , a standard Cauchy dispersion model, widely used for transparent dielectric thin films was applied and calibrated against reference samples of known thickness. The confidence interval of fitted lithium niobate thickness in these measurements was typically ± 0.3 nm. Measurements were carried out at the same location on the sample after each processing step, with thickness evaluated over a large rectangular region of interest to average across the area and suppress local fluctuations. Nevertheless, minor variations (~ 1 nm) may still occur due to the intrinsic surface topology of the sample and slight shifts in the measurement region. The observed results are presented in Fig. 4.

Figure 4 shows the evolution of LiNbO_3 film thickness with SC-1 treatment time. The blue square labeled as I represents the initial thickness of the LiNbO_3 thin film (Virgin Sample B), measured to be 320 nm, which closely matches the manufacturer specification. After a 5-minute

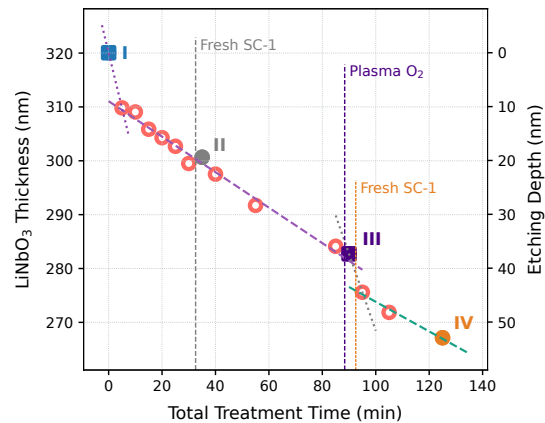


Fig. 4. Variation in the thickness of the LiNbO_3 thin film during SC-1 treatment. The film thickness was measured at regular intervals using spatially resolved ellipsometry. (I) Thickness of the virgin LiNbO_3 sample, (II) Fresh SC-1 solution, (III) Plasma O_2 treatment, (IV) SC-1 treatment after Plasma O_2 .

SC-1 treatment, a sharp decrease in thickness (approximately 10 nm) was observed (etch rate 2 nm/min). However, with subsequent treatment cycles, the etching rate stabilized and followed a slower linear trend (etch rate 0.33 nm/min). It is important to note that each thickness measurement required at least 20 minutes, during which the solution was kept on a hot plate and covered with a lid. Each time the sample was reintroduced into the SC-1 bath, stirring was temporarily stopped and the sample was placed gently.

To ensure the observed slower trend was not due to degradation of the SC-1 solution over time, a fresh batch was prepared, as described earlier, with confirmed concentrations. This is represented by the vertical gray dashed line in the Fig. 4. The next ellipsometric measurements confirmed that the observed linear trend was indeed due to the etching process, not the solution degradation, as no sharp decrease in thickness was observed.

The significantly faster etching rate during the first 5 minutes of SC-1 treatment suggests that a notable change occurs in the top 10 nm of the LiNbO_3 thin film. This is attributed to the existence of a here-termed "surface layer" whose composition differs from the remainder of the lithium niobate layer, being because of a different stoichiometry, of the presence of damages or of contaminations, and which is, irrespectively of its exact nature, misinterpreted as an additional layer of LiNbO_3 during the ellipsometry modeling.

In order to try to discriminate between these hypotheses, we exposed the sample for 5 minutes to an O_2 plasma under an O_2 flow rate of 1000 sccm, 1 mbar, and a 400 W RF power (Tepla plasma etcher) in order to induce some compositional changes [32] at the surface. This is indicated by the violet dashed line in the Fig. 4. The purple square, labeled as III, represents the measured thickness after plasma exposure.

Interestingly, the subsequent SC-1 treatment (5 minutes in a fresh SC-1 solution indicated by an orange dashed line on the figure) followed a similar trend then what we observed with the virgin surface (at I in Fig. 4), with a sharp increase in the etching rate that eventually stabilizes into a linear regime after repeated SC-1 treatments. The slopes of the two linear regimes before and after plasma exposure (etch rate 0.33 nm/min and 0.28 nm/min respectively), as shown in Fig. 4, remain consistent. We understand this results as a confirmation that a few nm thick damaged layer existed on the original sample, and is removed rapidly by SC-1 etching.

5. Post-SC-1 surface profilometry

The impact of SC-1 treatment on the sample surface topography was investigated using Atomic force microscopy (AFM) and ellipsometry.

The AFM measurements were performed at the center of the samples, covering a $2 \times 2 \mu\text{m}^2$ area to assess the surface roughness. A representative AFM image of the sample corresponding to label I in Fig. 4 is shown on the left of Fig. 5, revealing a root mean square (RMS) roughness of approximately 0.39 nm.

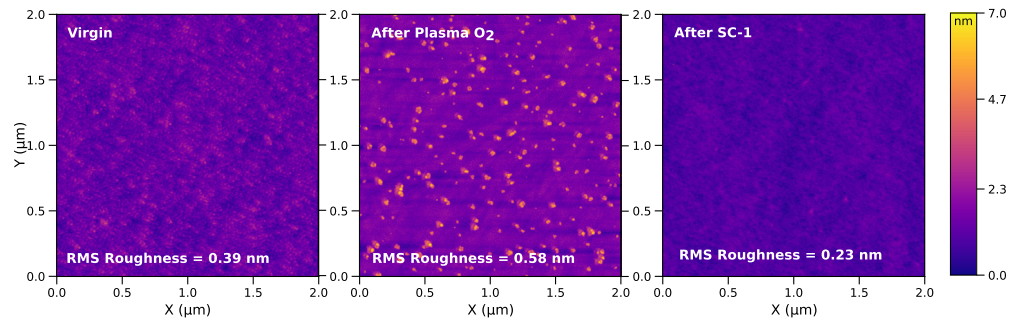


Fig. 5. Atomic force microscopy (AFM) images showing (Left) virgin, (Middle) O₂-plasma treated, and (Right) sequentially O₂-plasma and SC-1 treated samples, highlighting changes in surface roughness. The measured root mean square (RMS) roughness values are approximately 0.39 nm, 0.58 nm, and 0.23 nm, respectively.

The middle image in Fig. 5 shows the sample after O₂-plasma treatment (label III in Fig. 4). It reveals a significant increase in surface roughness and a high density of surface contaminants whose height is about 3–5 nm and whose lateral dimensions are in the tens-of-nanometers range. The latter observation compounded with the rapid increase in etching rate suggests that the plasma exposure is one but not necessarily the only way to create a surface layer at the top of the LiNbO₃ layer.

On the right of Fig. 5, the AFM image of the sample treated with SC-1 solution (label IV in Fig. 4) shows a significant reduction in surface roughness, decreasing from 0.58 nm after O₂-plasma treatment to 0.23 nm. This marks a 40 % reduction in roughness compared to the virgin sample on the left of Fig. 5, which initially had a roughness of 0.39 nm. These results indicate that the SC-1 solution effectively smooths the surface, restoring its morphology and making it more homogeneous.

To further assess the impact of SC-1 treatment, we evaluated the etching homogeneity over the full sample using ellipsometry measurements. The thickness of the LiNbO₃ thin film (Sample C) was measured at 15 evenly spaced points across the sample. The points were distributed over the sample area ($8 \times 12 \text{ mm}^2$) with an approximate spacing of 2 mm and positioned at least 2 mm away from the edges (Fig. 6 A, black dots), which are typically damaged during fabrication.

Such measurements were performed on the initial sample (Fig. 6 B), before the O₂-plasma treatment (Fig. 6 C) and at the end of process (Fig. 6 D). The results, presented in Fig. 6, show the change in surface profile relative to the sample center.

Each colored square represents an individual measurement, with an ellipsometry uncertainty of approximately 0.3 nm. The findings indicate no significant change in the sample homogeneity, suggesting uniform etching across the sample.

6. Surface layer composition influence

To gain insight into the nature of the surface layer, the chemical composition of the sample surface was analyzed using X-ray photoelectron spectroscopy (XPS) at key stages of the fabrication

process: virgin sample: before Si_3N_4 deposition (Sample D), after oxygen plasma treatment (Sample E), and following SC-1 treatment (Sample F) as listed in Table 1. The analysis was carried out on the basis that this surface layer is a damaged or contaminated LiNbO_3 layer.

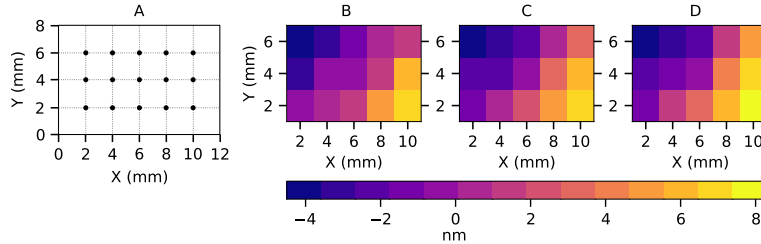


Fig. 6. (A) Measurement map and Surface homogeneity (B) the virgin surface, (C) after the initial SC-1 treatment, and (D) following a O_2 -plasma treatment and surface-restoration SC-1 process.

XPS measurements were performed using a K-Alpha spectrometer (Thermo Fisher Scientific) equipped with a monochromatic $\text{Al K}\alpha$ X-ray source (1486.6 eV). The binding energy scale was calibrated using the C_{1s} peak at 285 ± 0.1 eV as a reference. A built-in charge compensation system was employed, and the pressure in the analysis chamber was maintained at 2×10^{-7} mbar. High-resolution spectra were acquired with a step size of 0.1 eV and a pass energy of 30 eV, while survey spectra were recorded with a step size of 1 eV and a pass energy of 100 eV. The analysis spot size was 400 μm . Spectral fitting and quantitative analysis of elemental composition and chemical states were conducted using Avantage software. The uncertainty in elemental quantification was $\pm 5\%$. Figure 7 shows the deconvoluted Nb 3d and O 1s XPS spectra for the samples, which were well consistent with previous reports [33–41]

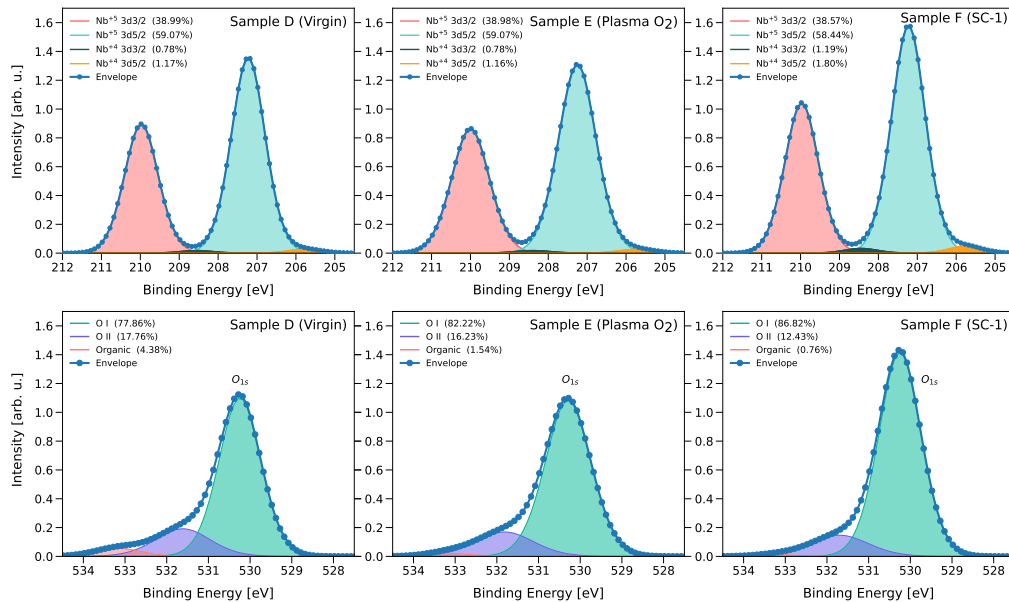


Fig. 7. Fitted and deconvoluted XPS spectra of (Top) Nb 3d and (Bottom) O 1s for (Left) sample D – virgin, (Middle) sample E – plasma O_2 treated, and (Right) sample F – SC-1 treated. The legend shows the percentage of each deconvoluted peak areas relative to the total envelope area for corresponding spectrum.

The binding energies for Nb 3d_{5/2} and Nb 3d_{3/2} are observed at 207.2 eV and 210.0 eV, as well as 205.8 eV and 208.5 eV, indicating the presence of both Nb⁵⁺ and Nb⁴⁺ oxidation states on the surface of LiNbO₃. The Nb⁵⁺ signals were significantly stronger than those of Nb⁴⁺, suggesting a predominance of Nb⁵⁺ ions. The intensity of Nb⁵⁺ peaks increased after SC-1 treatment, while the relative proportions of oxidation states remained mostly unchanged through out the samples. However, a slight relative increase in Nb⁴⁺ content was observed (+1 %), possibly due to NbO₂ precipitation or to the exposure of defect clusters formed during crystal growth [33], revealed after the surface layer was removed by SC-1 treatment.

The presence of Nb⁴⁺ indicates the formation of oxygen vacancies in the lattice for charge compensation [42]. To assess the impact of plasma O₂ and SC-1 treatments on lattice defects and oxygen vacancies, XPS spectra of the O 1s region were analyzed. The O 1s spectrum was asymmetric, shown in Fig. 7 (bottom), and was deconvoluted into three main components using Gaussian-Lorentzian fitting: O I at 530 eV (lattice oxygen), O II at 532 eV (oxygen vacancies), and a peak at 533.2 eV corresponding to organic species [35, 36, 38].

As shown in Fig. 7, the relative area of the O I peak increased from 77.86 % in the virgin sample to 82.2 % and 86.8 % after plasma O₂ and SC-1 treatments, respectively, indicating improved surface stoichiometry. These percentages are calculated relative to the total O 1s envelope for each sample. Notably, Sample E and F exhibited a higher absolute O I peak area compared to Sample D. This increase corresponds to a 7.6 % and 28.9 % rise in the O I peak area relative to Sample D. Concurrently, the organic component decreased from 4.38 % in Sample D to 1.54 % in Sample E and 0.76 % in Sample F, suggesting effective surface cleaning and removal of contaminants. These reductions represent 64 % and 80 % decreases, respectively, compared to the organic species peak in Sample D.

Additionally, the O II component, which reflects the concentration of oxygen vacancies, decreased from 17.79 % in Sample D to 16.23 % in Sample E and 12.43 % in Sample F. While these reductions may appear modest, they represent a 7 % and 19 % decrease in peak areas compared to Sample D, respectively, indicating a notable reduction in oxygen vacancies following the treatments.

The presence of oxygen vacancies means that there are voids in the surface layer, making it easier to etch during both dry and wet etching processes [23]. This is confirmed by the etching dynamics recordings shown in Fig. 4 where both the virgin and O₂-plasma-treated samples exhibit a higher etching rate compared to the sample treated with SC-1, which has fewer vacancies.

7. Modified fabrication protocol

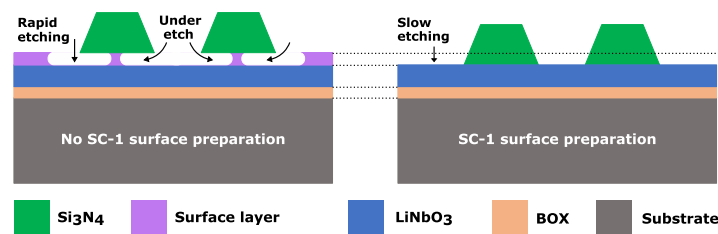


Fig. 8. Comparison of the layer stack for (Left) Sample A without SC-1 surface preparation, and (Right) Sample G with SC-1 surface preparation. In Sample A, the surface layer is constituted of LiNbO₃ with a higher concentration of oxygen vacancies, leading to fast under-etching. In contrast, the SC-1 treatment in Sample G removes this surface layer, resulting in a cleaner, restored surface with insignificant under-etching, suitable for Si₃N₄ deposition.

As a result of the above tests and analysis of the effect of SC-1 on thin film LiNbO_3 , we attribute our initial observation of Si_3N_4 grating peeling as an effect of the fast etching of the oxygen-vacancy-rich surface layer of the LNOI virgin samples. As depicted on Fig. 8 left, the SC-1 fast etching rate of the surface layer is also conducive to a strong under-etch of the Si_3N_4 grating wires, eventually leading to their peeling. On the contrary, provided that PECVD deposition of the Si_3N_4 layer does not re-introduce an equivalent degraded surface layer, its initial suppression with SC-1 should lead to structures with insignificant underetching (Fig. 8 right), and, in turn, provide a way to fabricate sub-micrometer-scale silicon nitride wires compatible with a final SC-1 cleaning of the surface.

We thus modified the fabrication process as outlined below, with changes highlighted in **bold**:

1. Cleaning:
Acetone/IPA: 5 minutes each
2. **Surface preparation using SC-1:**
 $\text{NH}_4\text{OH} : \text{H}_2\text{O}_2 : \text{H}_2\text{O} = 1 : 1 : 5$
Temperature: 65 °C, Stirring: 625 rpm
Duration: 30 minutes
3. Si_3N_4 deposition
4. Patterning
5. ICP etching
6. Resist stripping
7. Residual by-product cleaning using SC-1

Following the standard acetone/IPA cleaning, Sample G underwent SC-1 surface treatment for 30 minutes at 65 °C with continuous stirring at 625 rpm. A 70 nm layer of Si_3N_4 was then deposited via ICPECVD. The time between SC-1 treatment and Si_3N_4 deposition was kept minimal (≈ 15 minutes) to prevent contamination.

The standard fabrication protocol was then applied, with e-beam lithography and ICP etching.

Resist residues were removed using an acetone/IPA bath followed by a 5-minute oxygen plasma treatment. Finally, a second SC-1 treatment (5 minutes at 65 °C) was applied to remove any remaining by-products.

Figure 9 presents SEM images of the grating patterns in Sample A (left column) and Sample G (right column), highlighting the benefits of SC-1 surface preparation. The identical grating patterns, fabricated with and without SC-1 treatment, are directly compared. It is evident that in Sample A, which lacked the initial SC-1 treatment, the features have completely peeled off and coagulated. In contrast, the gratings in Sample G remain intact, with no visible signs of peeling or delamination, thereby confirming a significant improvement in adhesion between the Si_3N_4 and LiNbO_3 layers due to the SC-1 surface treatment. This surface preparation enabled the successful fabrication of patterns as narrow as 80 nm, which is near the resolution limit (~ 50 nm) of the electron beam lithography (EBL) writer. We anticipate that further optimization of the exposure dose could enable even finer patterning. Figure 9 does not include the patterns on Sample A corresponding to the $w = 80$ nm on Sample G (bottom right), as they were not developed properly due to issue during exposure.

Interestingly, wider features such as the 500 nm lines as shown in Fig. 3, the widest in our design remain intact even in Sample A. This observation suggests that wider patterns are inherently more resistant to peeling, which may also explain why this adhesion failure has not been more widely reported so far. As shown in Fig. 9, such effects are only apparent in narrower features, emphasizing the importance of surface preparation for nanoscale structures.

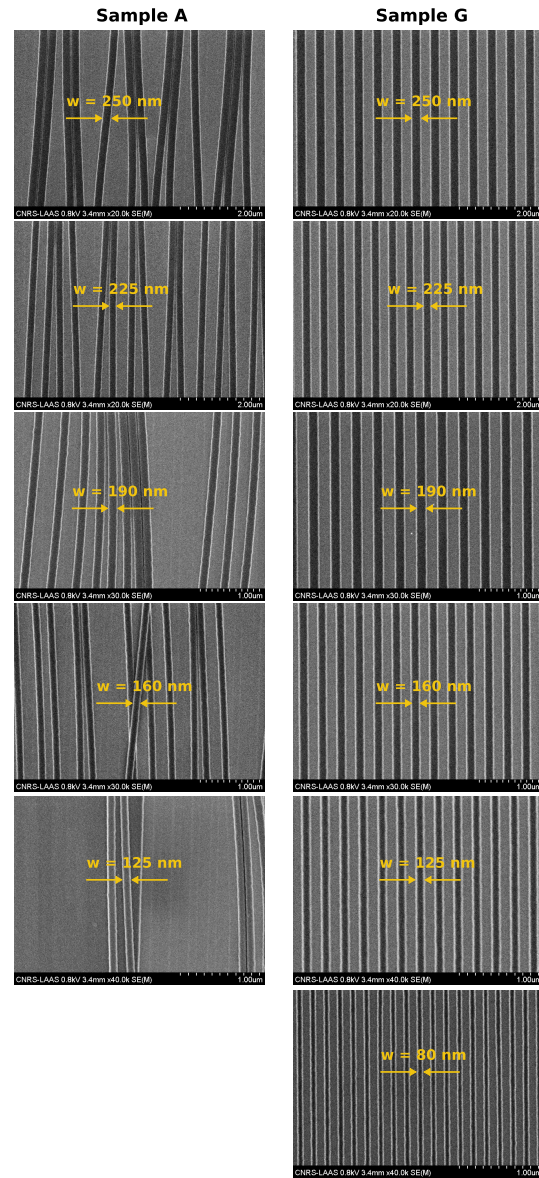


Fig. 9. SEM images of Si_3N_4 gratings with varying width (w), after residual by-product removal, for (Left column) Sample A – without SC-1 surface preparation, and (Right column) Sample G – with SC-1 surface preparation. Sample A shows significant peeling and coagulation, indicating poor adhesion. In contrast, Sample G exhibits uniform, well-defined features with no signs of delamination, confirming improved adhesion between the Si_3N_4 and LiNbO_3 films.

8. Conclusion

The above study establishes that the SC-1 cleaning process notably depends on the physical and chemical characteristics of LiNbO_3 thin films. More specifically, as evidenced by the reported etch kinetics and XPS measurements, the SC-1 treatment leads to the removal of the surface layer at an etch rate that is correlated to the level of oxygen vacancies present in the layer.

The understanding of this effect can be used in various ways when processing LiNbO₃-based devices. In particular, we have shown here that the SC-1 treatment can be performed prior to a Si₃N₄ deposition to reliably fabricate clean exfoliation-free structures whose critical dimensions can be smaller than 100 nm. Alternatively, one may consider using the afforded selective undertechning capabilities to implement the lift-off of sub-micron scale mask patterns on top of LiNbO₃ in other processes.

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